

8 (**short**) A viscoelastic material can be modelled using the spring-dashpot network shown in Fig. 7 .

(a) For this material, derive a differential equation relating the total applied stress σ , the total strain ϵ , time t and the material constants. [8]

(b) *Without further calculation*, sketch the equivalent spring-dashpot networks for this material when subjected to:

- (i) high frequency deformation, and
- (ii) low frequency deformation. [2]

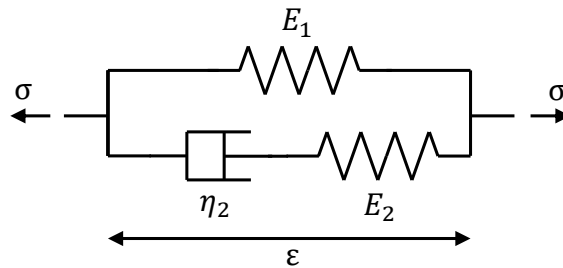


Fig. 7